

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900 - Columns: 18 - Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	1
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	1
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

1. AgeGroup Distribution – Find how many customers belong to each age group

age_group text	total_customers bigint
25-40	1188
41-55	1121
Above 55	1105
Under 25	486

2. Category-wise Revenue Analysis – Find the total and average purchase amount by each product category.

category text	total_revenue numeric	avg_purchase numeric
Clothing	104264	60.03
Accessori...	74200	59.84
Footwear	36093	60.26
Outerwear	18524	57.17

3. Seasonal Sales Performance – Find which season generates the highest sales revenue.

season text	total_orders bigint	total_revenue numeric
Fall	975	60018
Spring	999	58679
Winter	971	58607
Summer	955	55777

1. 4. Discount Impact on Spending – Find whether customers who use discounts spend more than those who don't.

discount_applied text	total_customers bigint	avg_purchase numeric
No	2223	60.13
Yes	1677	59.28

5. Most Popular Payment Methods – Find which payment methods are most frequently used by customers.

payment_method text	total_transactions bigint
PayPal	677
Credit Card	671
Cash	670
Debit Card	636
Venmo	634
Bank Transfer	612

6. Gender-wise Spending Pattern – Find the spending difference between male and female customers.

gender text	total_customers bigint	total_spent numeric	avg_spent numeric
Female	1248	75191	60.25
Male	2652	157890	59.54

7. Review Rating vs Purchase Amount – Find whether higher review ratings relate to higher purchase amounts.

review_rating double precision 🔒	total_customers bigint 🔒	avg_purchase numeric 🔒
[null]	37	66.27
5	68	64.35
4.9	162	63.75
4.8	144	61.88
4.7	148	59.28
4.6	170	57.31
4.5	139	59.49
4.4	158	60.53
4.3	147	59.67
4.2	169	60.76
4.1	148	61.96
4	181	59.24

8. Top 10 Highest Spending Customers – Find the customers who have spent the most overall.

customer_id bigint 🔒	age bigint 🔒	gender text 🔒	location text 🔒	total_spent numeric 🔒	total_orders bigint 🔒
2995	69	Female	Nevada	100	1
2215	50	Male	Idaho	100	1
1301	67	Male	West Virginia	100	1
3726	18	Female	Oklahoma	100	1
3322	32	Female	Maine	100	1
3838	45	Female	New Mexico	100	1
1406	33	Male	New Hampshi...	100	1
96	37	Male	Missouri	100	1
1457	59	Male	Montana	100	1
1848	46	Male	Arizona	100	1

9. Subscription Status vs Spending – Find whether subscribed customers spend more than non-subscribed customers.

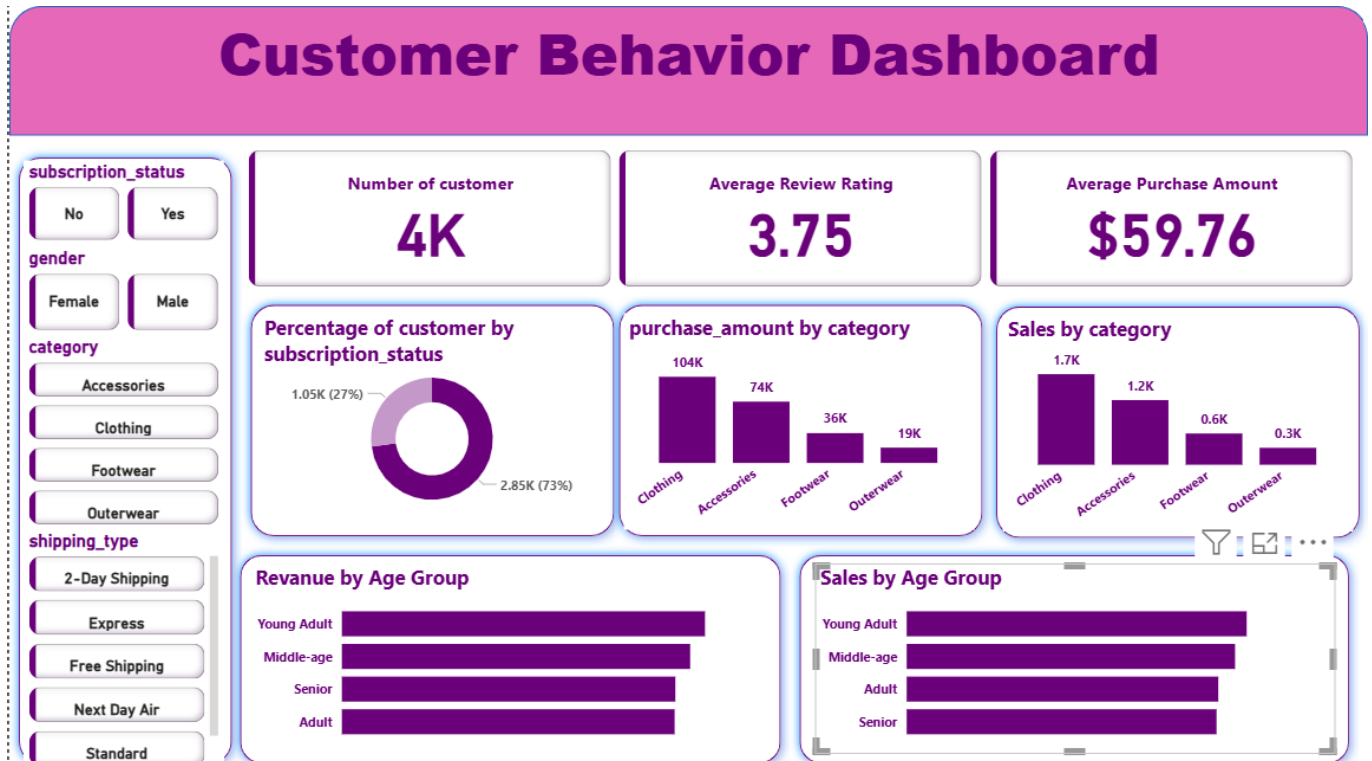
	subscription_status text 🔒	total_customers bigint 🔒	avg_purchase numeric 🔒	total_revenue numeric 🔒
1	No	2847	59.87	170436
2	Yes	1053	59.49	62645

10. Locations by Revenue – Find which locations generate the highest revenue for the company.

	location text	total_customers bigint	total_revenue numeric
1	Montana	96	5784
2	Illinois	92	5617
3	California	95	5605
4	Idaho	93	5587
5	Nevada	87	5514
6	Alabama	89	5261
7	New York	87	5257
8	North Dako...	83	5220
9	West Virgin...	81	5174
10	Nebraska	87	5172

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.